INCH-POUND

MIL-O-70688 (AR)
13 June 1990

MILITARY SPECIFICATION

OUTPUT CIRCUIT BOARD: 12594680

This specification is approved for use within the U.S. Army Armament,Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the pre-amplifier assembly 12594680 herein referred to as the Output Circuit Board, a sub-assembly of the Servo Electronics 10548193.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.2.1 Specifications and standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Material Manufacture and Inspection for General Specification
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-810	- Environmental Test Methods
MIL-STD-2000	- Standrad Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

MILITARY (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC).

10548193	- Servo Electronics
12594680	- Output Circuit Board Assembly

PACKAGING DATA SHEET

PS12594680	- Packaging of Output Circuit Board Assembly

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisitions functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets for MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12594680, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the output board of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

3.3 Environmental requirements.

3.3.1 Vibration (non-operating). Output Circuit Board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically. The cycle time shall be 120 minutes per axis. The sweep rate shall be 3.75 minutes ± 10% per decade. The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.3.2 Shock. The Output Circuit Board shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks, consisting of three shocks impulses of 15 ± 10% g's, 11 ± 10% milliseconds, half-sine wave applied in each direction along each of the three mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The Output Circuit Board shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Non-operating. The Output Circuit Board shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea leval (5.5 inches Hg).

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3.3.4 Temperature.

3.3.4.1 Operating. The Output Circuit Board shall meet the requirements of 3.4 and 3.5 when stabilzed thermally at any temperature between -40 degrees F and +155 degrees F.

3.3.4.2 Non-Operating (storage). The Output Circuit Board shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (Non-operating). The Output Circuit Board shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4. Electrical. Each conductor shall be electrically continuous, as specified on the schematic diagram of the Output Circuit Board Assembly.

3.5 Performance.

3.5.1 Operation. The Output Circuit Board shall meet the requirements of this operation while operating under test.

3.5.2 Regular voltage. The Output Circuit Board shall meet the following requirements:

3.5.2.1 Measure between 12.0 to 14.5 VDC for regulated voltage at terminal 10 with relation to terminal 2.

3.5.2.2 Measure between 12.0 to 14.5 VDC for regulated voltage for a power supply of 19.5 VDC and 30.5 VDC free of erratic or oscillatory behavior.

3.5.3 Reference voltage and frequency. The Output Circuit Board shall meet the following requirements:

3.5.3.1 Observe 35 to 45 VPP at Terminal 3 with relation to terminal 11.

3.5.3.2 For a resistance between 5K and 50K ohms the frequency of the voltage shall be between 1450 Hz and 1550 Hz.

3.5.4 Base drive. The following voltages at the base drives are full wave with no distortion, between Red and Black (Figure 1).

3.5.4.1 Measure the base voltage between +0.6 to 0.9 VDC.

3.5.4.2 Measure the base voltage between 0.75 to 1.25 VDC.

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3.5.4.3 Measure the base voltage between -0.75 to 1.25 VDC.

3.5.4.4 Measure the base voltage between +0.6 to 0.9 VDC.

3.5.4.5 Measure the base voltage between -0.75 to 1.25 VDC.
3.5.4.6 Measure the base voltage between -0.75 to 1.25 VDC.

3.5.4.7 Measure the base voltage between +0.6 to 0.9 VDC.

3.5.4.8 Measure the base voltage between -0.75 to 1.25 VDC.

3.6 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Responsibilty for compliance. All items must meet all requirements of Sections 3 and 5 and all applicable drawings. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsiblity of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 F to +90 F
b.	Relative Humidity	90% RH Max.
c.	Pressure	28 to 32 inches Hg

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) antenna assemblies for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may by subjected by the Government to any or all of the examinations and tests specified in Table I and Table II herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

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4.4.2 Examination and tests.

4.4.2.1 Components and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.3.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.3.2.3 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. The requirements and tests in Table I shall be inspected on a 100 percent basis.

TABLE I. Classification of Defects.

Characteristic		Requirements	Test Procedures
CRITICAL:		None	N/A
MAJOR:			
101.	Electrical	3.4	4.6.2
102.	Performance	3.5	4.6.3
MINOR:		None	N/A

4.5 Special sampling. One output board assembly out the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

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TABLE II. Special sampling.

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURES
CRITICAL:		None	N/A
MAJOR:			
103.	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3.3.3	4.7.1.3
106.	Temperature	3.3.4	4.7.1.4
107.	Humidity (non-operating)	3.3.5	4.7.1.5
MINOR:		None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the government until necessary corrections to all production since the last test and to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination and "Test Facilities" requirements of MIL-F-13926 wherever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials,

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dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.5.2.2, the standard test equipment listed in Table III shall be utilized in performing the applicable test of 4.6.

TABLE III. Test Equipment List.

Item Numbers	Description
1. Voltmeter, digital	0.1 mV to 100V, accuracy 0.1% ±1 digit (DC) 1.0% + 5 digit (2 to 5 KHz) (AC)
2. Oscilloscope	Input sensitivity 5 mV/Division Horizontal deflection 20ns/Division. Dual trace. Vertical amplifier response dc to 20 MHz

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only Steps 4 through 8 of paragraph II-2.1 apply. The antenna assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the antenna assembly shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

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d. The shock measuring system shall have a flat frequency response within 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the antenna assembly shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures I and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (see Table III) and visual examination for compliance with 3.4.

4.7.3 Performance. The Output Circuit Board shall be tested using the test set-ups as shown in Figures 1 (1.A). Apply 15 VDC ± 2% to the test fixture.

4.7.3.1 Regulator voltage. Verify compliance with Paragraph 3.5.2.

4.7.3.1.1 Verify compliance with paragraph 3.5.2.1 using digital voltmeter (Table III, item 1) by measuring at terminal 10 with relation to terminal 2.

4.7.3.1.2 Verify compliance with paragraph 3.5.2.2 by observing with the scope (Table III, item 2).

4.7.3.2 Reference voltage and frequency. Verify compliance with paragraph 3.5.3.

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4.7.3.2.1 Verify compliance with paragraph 3.5.3.1 using digital voltmeter (Table III, item 1) measuring at terminal 3 with relation to terminal 11.

4.7.3.2.2 Verify compliance with paragraph 3.5.3.2 by observing with the scope (Table III, item 2).

4.7.3.3 Base drive. Using digital voltmeter (Table III, item 1) verify compliance with paragraph 3.5.4.

4.7.3.3.1 Set S1 at position 1 and S2 at position Up. Verify compliance with paragraph 3.5.4.1.

4.7.3.3.2 Set S1 in position 2 and S2 at position Up. Verify paragraph 3.5.4.2.

4.7.3.3.3 Set S1 at position 3 and S2 at position Up. Verify compliance with paragraph 3.5.4.3.

4.7.3.3.4 Set S1 at position 4 and S2 at position Up. Verify compliance with paragraph 3.5.4.4.

4.7.3.3.5 Set S1 at position 4 and S2 at position Down. Verify compliance with paragraph 3.5.4.5.

4.7.3.3.6 Set S1 at position 3 and S2 at position Down. Verify compliance with paragraph 3.5.4.6.

4.7.3.3.7 Set S1 at position 2 and S2 at position Down. Verify compliance with paragraph 3.5.4.7.

4.7.3.3.8 Set S1 at position 2 and S2 at position Down. Verify compliance with paragraph 3.5.4.8.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

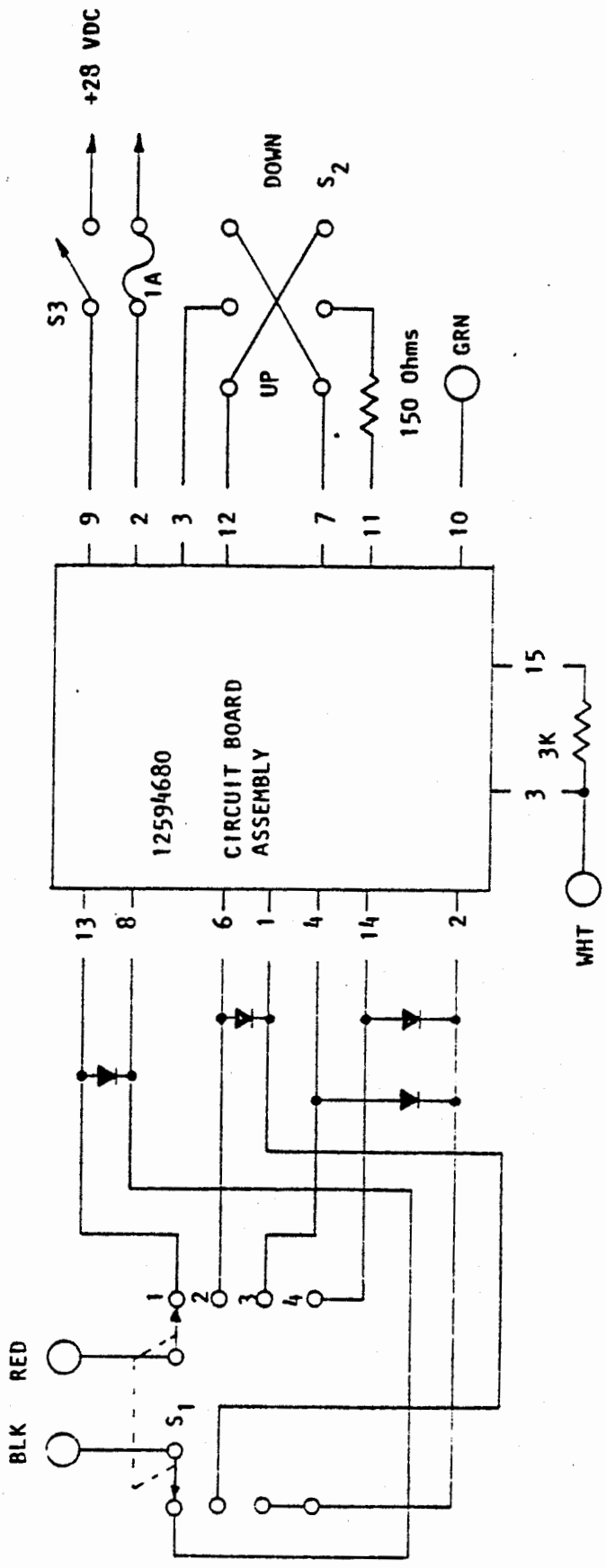
5.1 General. Preservation, packaging, and marking shall be in accordance with Packaging Data Sheet PS 12594680. The level of protection shall be defined in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Output Circuit Board converts AC modulated output from Driver Board to DC drive signal required to drive the output bridge transistors. It also contains a regulated power supply and power carrier frequency oscillator circuit.

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NOTE: Apply 28 VDC ± 2% to the text fixture unless otherwise specified.

FIGURE 1

Output Circuit Board Test Set-up

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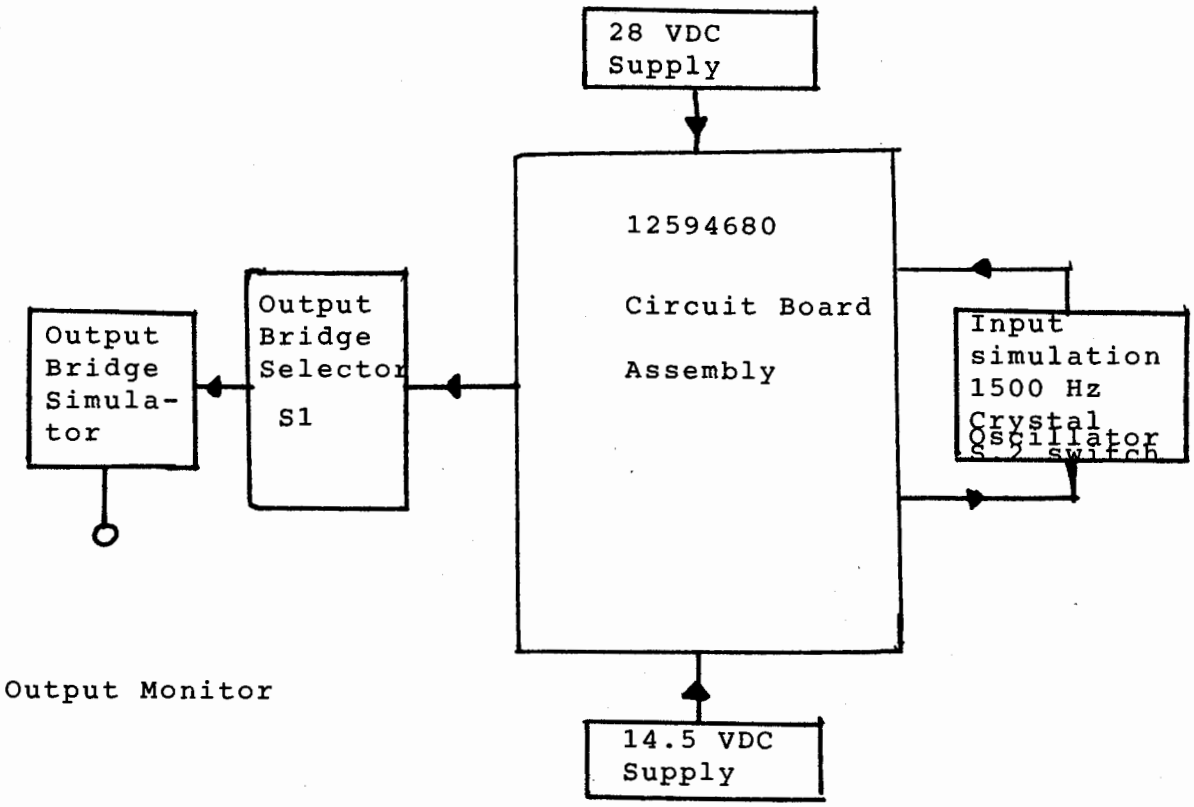


Figure 1A

Output Circuit Board Test Set-up

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6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First Article Sample Requirements.
d. Packaging Requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of Conformance for each lot or shipment of product.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-OT-90-12009 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Radar Set AN/VPS-2
Servo Electronics
Vulcan Air Defense System (VADS)

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A769)

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